clinical failure: fixed-effect meta-analysis (odds ratio)

Comparison: Intervention / Control

Methods

Odds ratios were pooled using a fixed-effect model. A continuity correction of 0.5 was applied to individual study estimates only; the pooled Peto estimate uses the original counts. Heterogeneity was assessed with Cochran's Q and I^2. Analyses were performed in RC MetaStudio 0.5.1. R packages loaded for this analysis: metafor 5.0.1, meta 8.5-0.

Results

19 studies were included (1097 participants in Intervention, 1080 in Control). The pooled odds ratio was 0.710 (95% CI [0.541, 0.930], z = -2.482, p = .013). The confidence interval excludes 1, with lower odds of the outcome in Intervention than in Control. Heterogeneity: Q = 41.000, df = 18, p = .002. I^2 = 56.1%, which may represent moderate to substantial heterogeneity (Cochrane Handbook ranges).

Table 1. Pooled effect

Effect	Estimate	95% CI	SE	z	p
Odds ratio	0.710	0.541 to 0.930	0.138	-2.482	.013

Table 2. Model fit

Statistic	Value
Studies (k)	19
Participants in Intervention	1097
Participants in Control	1080
Cochran’s Q	41.000
Heterogeneity degrees of freedom	18
Heterogeneity p	.002
I^2	56.1%

Table 3. Study results

Study	Effect	95% CI	Weight
Gonzalez	0.581	0.178 to 1.889	5.3%
Prins	0.460	0.127 to 1.668	4.4%
Giamarellou	0.163	0.031 to 0.865	2.2%
Maller	1.394	0.672 to 2.894	13.8%
Sturm	0.254	0.026 to 2.519	0.9%
Marik	0.405	0.240 to 0.682	26.9%
Muijsken	1.597	0.496 to 5.148	5.4%
Vigano	2.592	0.160 to 41.886	0.5%
Hansen	0.198	0.058 to 0.671	4.9%
De Vries	1.914	0.500 to 7.326	4.1%
Mauracher	0.172	0.038 to 0.775	2.8%
Nordstrom	0.669	0.139 to 3.220	3.0%
Rozdzinski	1.018	0.465 to 2.229	12.0%
Ter Braak	1.352	0.504 to 3.624	7.6%
Tulkens	1.000	0.020 to 50.397	0.0%
Van der Auwera	0.252	0.025 to 2.516	0.9%
Klastersky	4.738	1.396 to 16.084	4.9%
Vanhaeverbeek	1.050	0.021 to 52.979	0.0%
Hollender	0.352	0.022 to 5.700	0.5%

clinical failure · Binary · Odds Ratio · Comparison: Intervention / Control

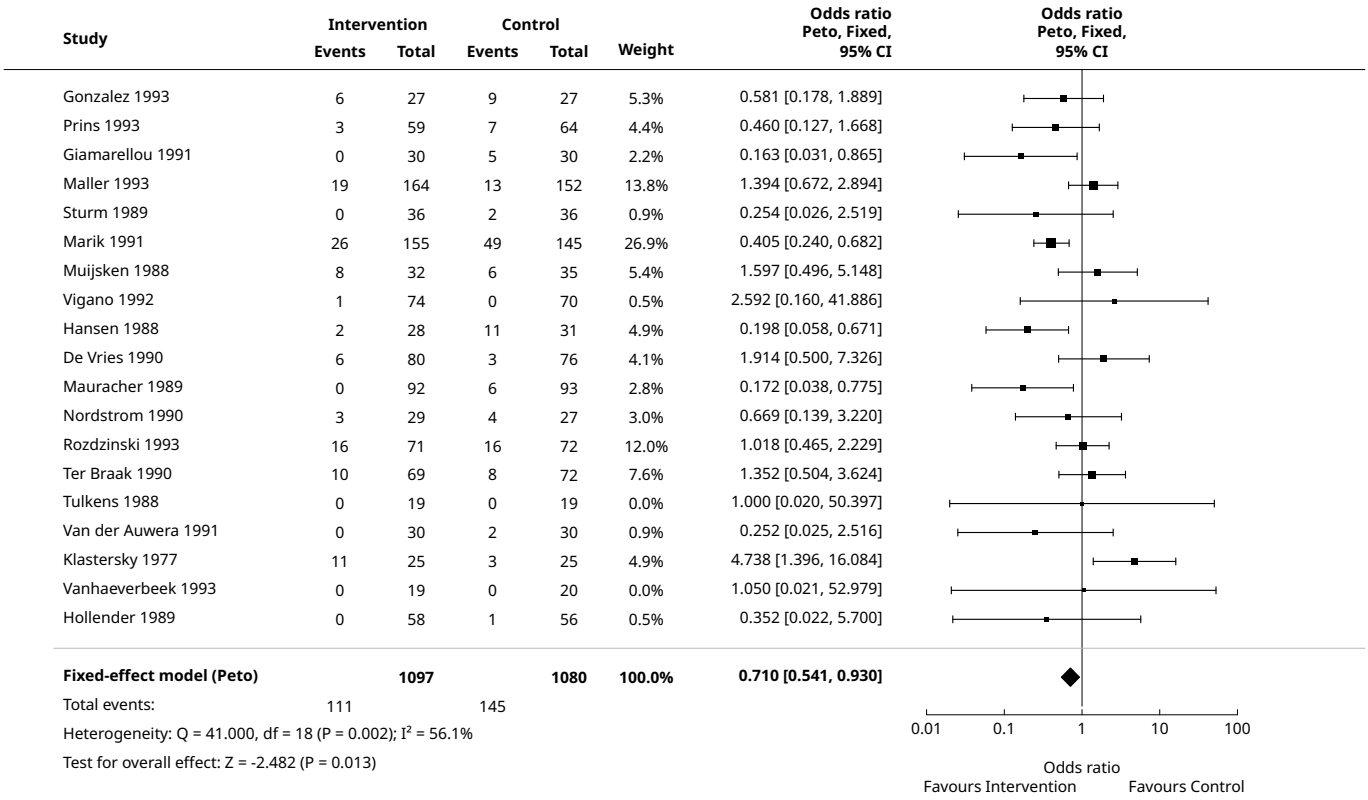

Figure 1. Forest Plot

Supplementary details

Table 4. Full-precision results

Source	Results
Summary — Model Results	Estimate: 0.710 Lower bound (95% CI): 0.541 Upper bound (95% CI): 0.930 p-value: 0.013
Summary — Heterogeneity	Q: 41.000 df: 18 p-value: 0.002 I²: 56.1%
Summary — Test for overall effect	Z: -2.482 p-value: 0.013
Summary — Calculation scale (log)	Estimate: -0.343 Lower bound (95% CI): -0.614 Upper bound (95% CI): -0.072 SE (log scale): 0.138
Weights — Study weights	Study: Gonzalez Effect: 0.581 Lower bound: 0.178 Upper bound: 1.889 Weight (%): 5.3%
Weights — Study weights	Study: Prins Effect: 0.460 Lower bound: 0.127 Upper bound: 1.668 Weight (%): 4.4%
Weights — Study weights	Study: Giamarellou Effect: 0.163 Lower bound: 0.031 Upper bound: 0.865 Weight (%): 2.2%
Weights — Study weights	Study: Maller Effect: 1.394 Lower bound: 0.672 Upper bound: 2.894 Weight (%): 13.8%
Weights — Study weights	Study: Sturm Effect: 0.254 Lower bound: 0.026 Upper bound: 2.519 Weight (%): 0.9%
Weights — Study weights	Study: Marik Effect: 0.405 Lower bound: 0.240 Upper bound: 0.682 Weight (%): 26.9%
Weights — Study weights	Study: Muijsken Effect: 1.597 Lower bound: 0.496 Upper bound: 5.148 Weight (%): 5.4%
Weights — Study weights	Study: Vigano Effect: 2.592 Lower bound: 0.160 Upper bound: 41.886 Weight (%): 0.5%
Weights — Study weights	Study: Hansen Effect: 0.198 Lower bound: 0.058 Upper bound: 0.671 Weight (%): 4.9%
Weights — Study weights	Study: De Vries Effect: 1.914 Lower bound: 0.500

	Upper bound: 7.326 Weight (%): 4.1%
Weights — Study weights	Study: Mauracher Effect: 0.172 Lower bound: 0.038 Upper bound: 0.775 Weight (%): 2.8%
Weights — Study weights	Study: Nordstrom Effect: 0.669 Lower bound: 0.139 Upper bound: 3.220 Weight (%): 3.0%
Weights — Study weights	Study: Rozdzinski Effect: 1.018 Lower bound: 0.465 Upper bound: 2.229 Weight (%): 12.0%
Weights — Study weights	Study: Ter Braak Effect: 1.352 Lower bound: 0.504 Upper bound: 3.624 Weight (%): 7.6%
Weights — Study weights	Study: Tulkens Effect: 1.000 Lower bound: 0.020 Upper bound: 50.397 Weight (%): 0.0%
Weights — Study weights	Study: Van der Auwera Effect: 0.252 Lower bound: 0.025 Upper bound: 2.516 Weight (%): 0.9%
Weights — Study weights	Study: Klastersky Effect: 4.738 Lower bound: 1.396 Upper bound: 16.084 Weight (%): 4.9%
Weights — Study weights	Study: Vanhaeverbeek Effect: 1.050 Lower bound: 0.021 Upper bound: 52.979 Weight (%): 0.0%
Weights — Study weights	Study: Hollender Effect: 0.352 Lower bound: 0.022 Upper bound: 5.700 Weight (%): 0.5%

Pooled Odds Ratio: 0.710 (95% CI 0.541 to 0.930).

References

1. Peto one-step odds-ratio method: Yusuf, S., Peto, R., Lewis, J., Collins, R., & Sleight, P. (1985). Beta blockade during and after myocardial infarction: An overview of the randomized trials. Progress in Cardiovascular Disease, 27(5), 335-371. doi:10.1016/S0033-0620(85)80003-7.

2. Sparse-data comparison: Bradburn, M. J., Deeks, J. J., Berlin, J. A., & Localio, A. R. (2007). Much ado about nothing: A comparison of the performance of meta-analytical methods with rare events. Statistics in Medicine, 26(1), 53-77. doi:10.1002/sim.2528.

3. Implementation reference: Viechtbauer, W. (2010). Conducting meta-analyses in R with the metafor package. Journal of Statistical Software, 36(3), 1-48. doi:10.18637/jss.v036.i03.